

Appendix 7: Common Problems and Overhaul

Instruction Manual

Instruction Manual Version: V2.05

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Chapter 1 Frequently Asked Questions and Overhaul

1、In Accordance with the Instructions to Install the USB Driver, the Device Manager Appears Device Name, But the Yellow "!" or "?" or "?" Warning Symbol.

Solution: The driver has been installed normally, you need to restart the system to complete the installation and initialization of the USB driver.

If the problem is not solved after reboot, it means that the driver installation step is wrong, and you need to uninstall the driver and reinstall it.

2、Under the USB_CAN Tool Software, When Starting the Device, It Prompts "No USB-CAN Device Found".

Solution: The possible reason for this failure is that the USB_CAN adapter driver is not properly installed. Follow the instructions to install the USBCAN device driver.

3、Under Device Manager, the USB Device Name Is Not Consistent With the Manual or Model Number.

There is only one kind of USB driver, WinUSB, the name of which will appear according to different systems and models: USBCAN Device, CANalyst-II Device, WinUSB Device, etc. The names are common, and it is normal for any of them to appear.

4、Is It Always Necessary to Use 120Ω Termination Matching Resistors?

Two 120Ω termination resistors must be guaranteed on the bus, otherwise it will affect the normal operation of the CAN bus.

The 120Ω termination matching resistors are used to absorb endpoint reflections and provide a stable physical link. Please ensure that the bus is connected with 120Ω termination resistors at both ends.

The USB_CAN adapters (USBCAN-2A, USBCAN-2C Supreme Edition) have a

built-in 120Ω termination resistor for each channel. If you need to connect a termination resistor when networking, you only need to short the termination resistor terminals (R+,R-) of the corresponding CAN channel with a wire.

The CANalyst-II analyzer has two 120Ω termination resistors for each channel, one for each dip switch. If there is a need to connect the termination resistors during networking, users only need to turn the corresponding resistor dip switches down to the ON position.

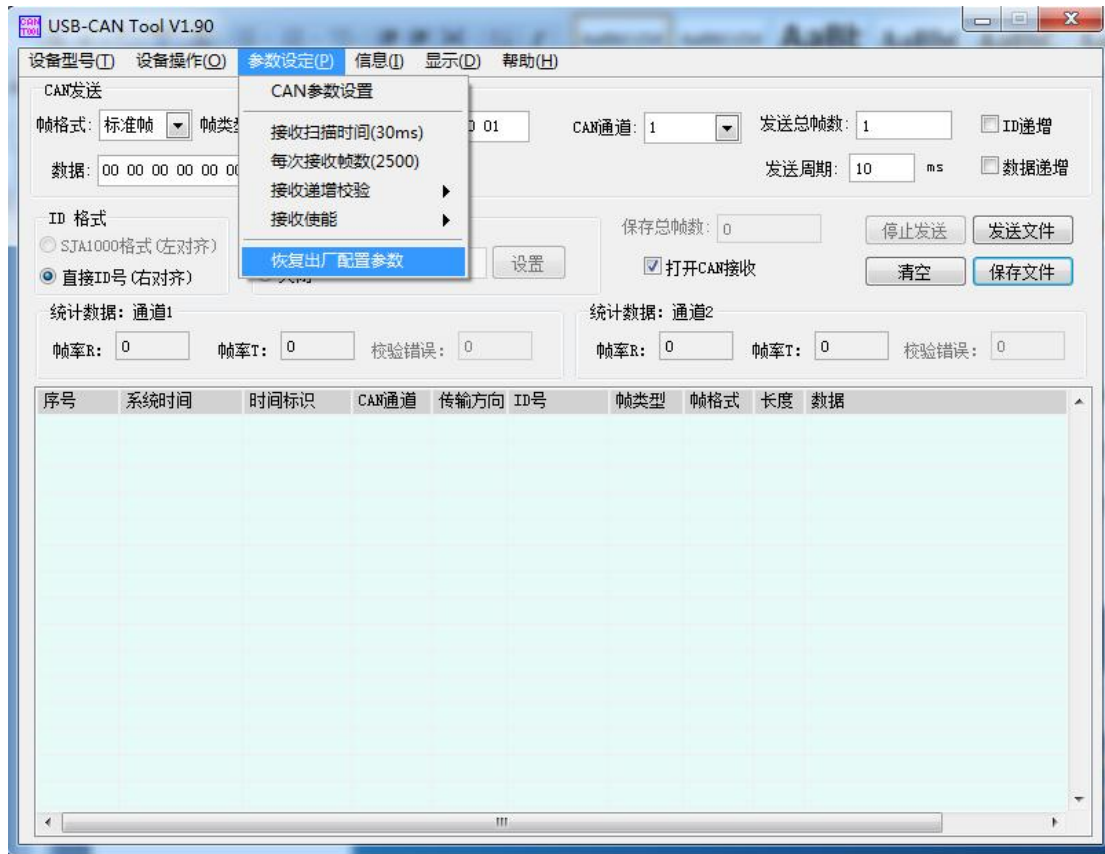
① If other nodes on the bus have two 120-ohm termination resistors, the USB-CAN adapter does not need to access the built-in 120-ohm termination resistor;

② If other nodes on the bus have one 120-ohm termination resistor, then short the R+ and R- of the corresponding CAN port of the USB-CAN adapter, or flip down one of the dip switches of the corresponding CAN port of the CANalyst-II Analyzer Supreme Edition, and connect the built-in resistor to the bus;

③ If other nodes on the bus do not have 120-ohm terminating resistors, then after shorting R+ and R- of the corresponding CAN port of the USB-CAN adapter to connect the built-in resistor to the bus, it is also necessary to hang an external 120-ohm terminating resistor on top of the bus. Or toggle the two dip switches of the corresponding CAN ports of the CANalyst-II Analyzer Supreme Edition.

In the case of bus connectivity, turn off the power of all devices on the bus, and test whether the resistance between CANH and CANL is 60 ohms, around 60 ohms indicates that the bus termination resistor is configured normally.

5、In The Use of the Process, Has Successfully Received Data, but Modify the Configuration, Can No Longer Receive Data, It Is Recommended to Restore the Debugging Software USB_CAN TOOL Factory Settings, You Can Directly Clear All the Incorrect Configuration.



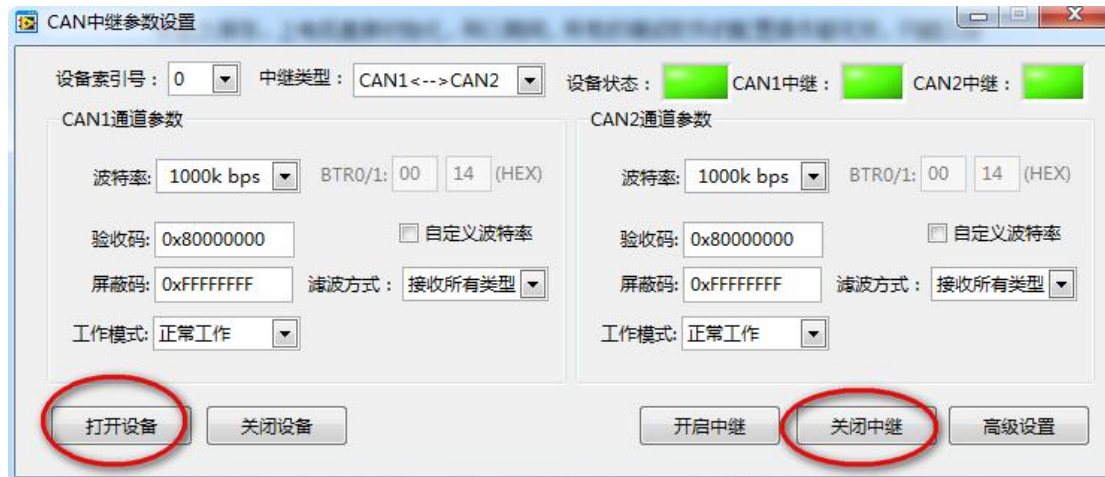
6 、 In the Use of the Process, Has Successfully Received Data, But After Modifying the Configuration, No Longer Receive Data, According to the Instructions to Do the Test Between the Two Channels to Send and Receive, Only Send a Frame, the Corresponding Channel Does Not Stop Receiving.

Solution: This is due to the fact that the relay mode was not turned off with the relay software after it was turned on. The so-called relay function is to realize the data exchange between two CAN buses by only supplying power to the USBCAN device after detaching from the PC. Relay function must be turned on and off through the relay configuration software to achieve, because the configuration data in the opening of the relay, will be cured in the device is permanently saved, power on directly after the initialization, so during the period, all the debugging software configuration operations are invalid, can only be sent and received with the current parameters of the data, i.e., transparent transmission function. You can only turn off the relay function through the relay configuration software, the relay configuration parameters are erased, and the relay mode exits and enters the normal mode.

So in relay mode, the baud rate can not be configured and no data can be received.

According to the manual to do the test between the two channels, only send a frame, the corresponding channel does not stop receiving because, CAN1 send a frame of data to CAN2, CAN2 received data, the relay function makes it transfer the data to CAN1 issued by CAN1 sent

through the CAN bus to CAN2, and so on and so forth.



7、Baud Rate Can Not Be Detected

Solution: The principle of baud rate detection is to continuously change the baud rate of the USBCAN device in the case of data transmission on the bus, during which the receiving error flag and receiving interrupt flag are constantly monitored until the baud rate is matched. So the premise of baud rate detection is CAN bus physical link is normal, that is, wiring, terminal resistance and other correct connection, and then there must be data sent above the bus, that is, the nodes above the bus must take the initiative to send data to the top of the bus.

8、Baud Rate Detection Result Is a Range, Which Value Should Be Taken.

Solution: baud rate deviation can receive data, but may be unstable, the international standard is 1% range can be that, 250K set $250 \pm 1\% * 250K$ can be. But 250K around other baud rates can also be received. So the baud rate detected is a range. So you want to get the most accurate baud rate, with a little experience, is generally detected in the middle of the range. Here the detection result is 200K-296.296K, so 250K is the most accurate.



9、During the Process of Using, the Baud Rate and Other Configuration Parameters Can Not Be Saved.

Solution: The configuration file cannot be generated due to some security settings. Right-click the USB_CAN TOOL shortcut, select "Run as administrator", so that the USB_CAN TOOL installation directory can generate the configuration file. Next time, just run the software directly without repeating the above steps.

10、According to the Manual to Do Two Channels to Send and Receive Each Other Test Is Normal, But Access to the Target Bus or Device, Can Not Read the Data.

Solution: The mutual transceiver test between two channels is normal, which means that the software and hardware of the debugger are completely normal. No need to suspect that the debugger problem. To check the following items one by one.

- ① CAN is a differential level, wiring only CANH, CANL two wires, to ensure

that the debugger and the bus wiring is correct: CANH to CAN high, CANL to CAN low. (CANalyst-II analyzer Supreme Edition more than one S terminal for the shielded wire wiring and lightning ground, for auxiliary wiring.)

② Ensure that there are two 120-ohm termination resistors across the bus after the debugger is connected to the bus.

③ Make sure the relay function is not turned on or is turned off. If it is not turned off, turn off the relay function according to the method mentioned earlier.

④ Make sure the debugging software is in the factory state, if not, please delete the configuration file and restore the factory settings according to the method mentioned earlier.

⑤ With the software in the factory state, start the device and configure the baud rate in the pop-up configuration dialog box (select directly from the baud rate drop-down list). Other parameters use the default values and do not change them. Under normal circumstances, please configure the baud rate when you start the device, do not enter the parameter settings to configure the baud rate.

⑥ Ensure that the device sends out signals: use an oscilloscope to probe the voltage between the high and low lines of the CAN bus, with the oscilloscope probe connected to CAN high and the oscilloscope ground connected to CAN low. Make sure that there is a waveform and the level is normal.

⑦ Ensure that the actual value of the device baud rate is consistent with the baud rate configured by the debugger.

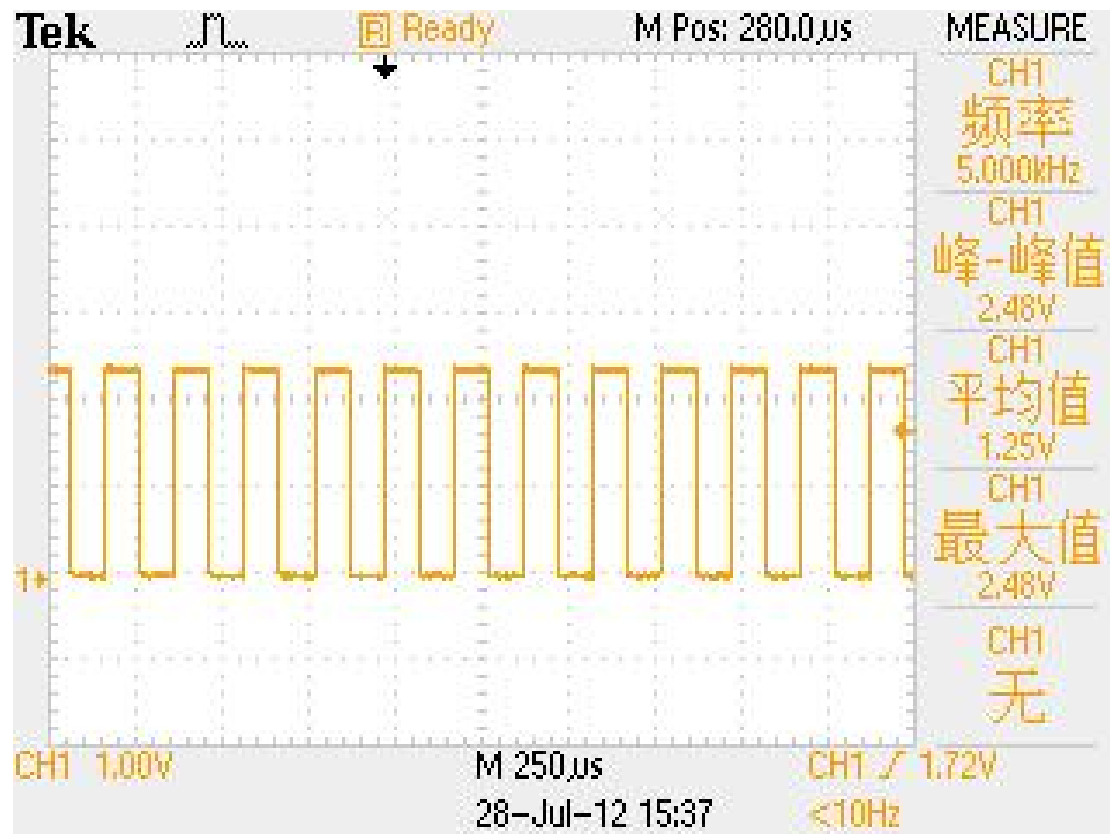
For standard CAN devices, i.e., finished devices purchased. Generally, the device can operate normally, and provide the corresponding baud rate value, directly configure the corresponding value can be. You can also use an oscilloscope to test the baud rate value.

For the development stage of the CAN device, such as debugging development boards, development of early devices. First of all, ensure that the hardware is normal: waveform output, normal level, and secondly, ensure that the actual value of the baud rate is consistent with the set value. Different chips have different setting methods. If the baud rate is not the set value, you need to refer to the chip instructions to

reconfigure.

How to measure the baud rate value with an oscilloscope: If you can control the device to send out any data, then let the device send out, ID and data are 10 or 01 alternately.

As shown in the figure below, the frequency of the square wave is 5kHz, which corresponds to a baud rate of 10 K. The differential level voltage is about 2V, which is slightly higher than normal.



11、The Debugger Sends Only One Frame, But the Data Is Always Displayed On the Bus.

Solution: In order to ensure that the sent data is not lost, the USB-CAN device has an automatic retransmission function, that is, when the USB-CAN device is not connected to the CAN bus, the baud rate does not match the CAN bus, the wiring is incorrect, or the termination resistor does not match, and the USB-CAN device can not receive the answer signal, then the USB-CAN device will automatically retransmit until the data is received by the other nodes of CAN bus or the USB-CAN device is

powered down and restarted. The USB-CAN device will automatically retransmit until the data is received by other nodes of the CAN bus or the USB-CAN device is powered down and restarted. Each channel has 1000 frames data buffer capacity, when the buffer 1000 frames is not full, the host computer will call the send function and return 1 to indicate that the data has been normally sent to the debugger buffer, waiting for sending. When the buffer is full of 1000 frames, the host computer will call the send function to return 0, indicating that the sending has failed, until the buffer data is sent normally.

There are several situations in which data is not sent out:

① Debugger configuration is incorrect. Please restore the factory settings according to the manual, and redo the test of sending and receiving between two channels.

Baud rate mismatch. Please test the baud rate of the debugger and the device according to the method mentioned above.

③ Wiring error. Please make sure CANH and CANL are wired correctly.

④ Termination resistor mismatch. Please make sure there are two terminal resistors above the bus.

⑤ Device on the bus is abnormal. Please make sure the equipment on the bus is working normally, which can be tested by testing the baud rate with an oscilloscope.

⑥ The device on the bus is in listen mode. The condition for the debugger to retransmit is that it does not receive an acknowledgement signal from the target device, i.e., an acknowledgement frame. If all the devices on the bus are in the listening mode, the devices on the bus can receive the data sent by the debugger but will not answer, and the debugger will automatically retransmit if it does not receive the answer signal. Please make sure that all devices on the bus are in normal operation mode.

12. Can the Device Be Used for Low Speed CAN Networks?

Solution: The CAN transceiver used in the USB_CAN adapter/CANalyst-II analyzer is the high-speed CAN transceiver TJA1040T. This series of transceivers has

the best versatility and can be used in any high-speed CAN network.

Low-speed fault-tolerant or single-wire CAN is only different in the physical layer, the CAN protocol is the same. Ultimately, the CAN transceiver is not the same as our CAN transceiver is TJA1040T high-speed CAN transceiver. Theoretically, it is not compatible with low-speed fault-tolerant or single-wire CAN.

However, some customers are able to use it for single-wire CAN. That is, connect CANH to CANH and CANL to GND. It is possible to communicate. Although theoretically it is not possible to use it in this way, in practice it is possible to communicate. You can try this method to ensure the safety of your person and property. **We are not responsible for any consequences of trying this method.**

There is no equipment in the market for debugging low-speed fault-tolerant or single-wire CAN. The best way to debug low-speed fault-tolerant or single-wire CAN with our products is to replace the CAN transceiver by yourself. **Since the replacement of CAN transceiver must damage the label and change the components, so we do not warranty.**

13、Secondary Development Program or CANtest, CANpro Software in the Case of Our Dynamic Link Library, In Some Computers Above the Running Error.

Solution: As shown in the figure below, the error: the application can not start normally 0xc0150002. download and install the VC2008 runtime library vcredist_x86 will be fine.



14. Can More Than One USB_CAN Adapter Be Installed In One Computer.

Sure, up to 50 or more USB_CAN adapters can be connected to one computer at the same time, and 50 adapters can be turned on individually by the USB-CAN Tool debugging tool. For the application related to the Multi-CAN, please refer to

Appendix 5: Multi-CAN Operation Manual for the description of the document.

15、What Is the Highest Data Conversion Rate of USB_CAN Adapter?

The maximum transmitting and receiving speed of single CAN channel of USB_CAN adapter is over 8000 fps, the frame mentioned here refers to data frame. If it is remote frame it may be faster. And two channels working at the same time can also reach more than 8000 fps per channel. In addition, the maximum data traffic will be affected by the performance of PC.

16、Why Is the CAN Status Indicator Light Not On?

Because all the operations of the USB_CAN adapter are controlled by the PC, only the PC sends a command to start the CAN communication, the CAN status indicator will be meaningful.

17、Why Does the System Operate Illegally When Calling the Interface Function?

First of all, when using the interface function, please read the function description carefully to ensure that the input parameters are legal, pay special attention to the transfer of pointers (address), or refer to the example program provided. If the problem is still not solved, you can contact our technical support.

18、How to Set the Baud Rate of USB_CAN Adapter?

In normal circumstances, please configure the baud rate when you start the device. Don't enter the parameter setting to configure the baud rate. The baud rate in the parameter setting is configured as advanced mode, which is generally for the case of very long transmission line, and the propagation delay needs to be corrected.

For secondary development, generally use Timing0 and Timing1 (parameters in the VCI_INIT_CONFIG structure) in the VCI_InitCan function to configure the baud rate (the value corresponding to the regular baud rate, you can find the corresponding cross-reference table in the instruction manual); rarely, customize the settings by

calling the SetReference2 function to set the baud rate of the USB_CAN adapter. USB_CAN adapter parameters, our devices support simple and professional customization and other baud rate settings, specific setup procedures, please refer to *"Annex 2: CAN parameter setting instructions"* document.

19、What Should I Pay Attention to When I Turn on And off the Device?

USB_CAN adapter provides two CAN ports, users can operate different ports in the same program. USB_CAN adapter does not allow sharing to open the device, and the same device can not be opened several times by different processes by calling VCI_OpenDevice function. VCI_OpenDevice and VCI_CloseDevice functions are usually called only once when the application initializes and exits. When closing the device, if the current port is no longer in use, you should first call VCI_ResetCAN function to make the current port off the CAN bus.

20、How to Use Interrupt to Operate the Communication Card?

USB_CAN adapter does not provide an interface to operate interrupts directly, because interrupts are already handled in the driver. Most of the reasons for operating interrupts in an application program are: the program doesn't know when the data will reach the device, and it needs to get triggered by a receive message in order to read the data from the buffer. The general means of solving this problem is to use multithreading (or multitasking). Start a new thread, call VCI_GetReceiveNum function cyclically in the thread to query whether there is data in the USB_CAN adapter receive buffer. If there is data, then call VCI_Receive function to read the corresponding data or directly call VCI_Receive to query the receive buffer and receive the corresponding data, the length of the received data should be set to a long time, the efficiency of this program is higher. In addition, the VCI_Receive function takes more than 5ms, so pay attention to control the interval between two calls to the VCI_Receive function in more than 5ms.